School of Computer Science Engineering and Information Systems

Department of Software and Systems Engineering

Fall Semester – 2026 - 27

SWE3004 - Software Design and Development Project

0th Review

Register Number	22MIS0630
Student Name	VARUN A
Project Domain	AI-Powered Smart Logistics and Sustainable Transportation
Project Title	AI-Driven Smart Logistics Platform for Warehouse Optimization, Delivery Safety, and EV Charging Management
Abstract	*The rapid expansion of quick-commerce and electric vehicle (EV)-based delivery services has introduced significant operational challenges across warehouse management, delivery operations, and sustainable transportation. Quick-commerce companies struggle with inefficient product picking inside dark stores, poor order batching that increases delivery time and operational costs, and unexpected demand surges during festivals, rainy weather, weekends, and peak hours. At the same time, delivery partners often take unsafe routes or exceed speed limits to meet strict delivery deadlines. As delivery fleets increasingly adopt electric vehicles, another major challenge is the lack of real-time charging station availability, resulting in long waiting times, route disruptions, and reduced delivery efficiency.* *The proposed system develops an **AI-Driven Smart Logistics Platform** that integrates intelligent warehouse optimization with delivery fleet management. For warehouse operations, the platform provides AI-powered indoor navigation that guides workers through optimized picking sequences, reducing product search time and improving order fulfillment efficiency. Intelligent order batching*

	algorithms group nearby customer orders while considering delivery deadlines, traffic conditions, and rider availability. Machine learning models forecast peak-hour demand using historical orders, weather forecasts, holidays, and local events, enabling proactive inventory positioning and workforce allocation before demand surges occur. *For delivery operations, the platform dynamically adjusts promised delivery times based on live traffic and weather conditions, reducing pressure on delivery partners and encouraging safer driving practices. Instead of rewarding delivery speed alone, the system introduces a* ***Safety Score*** *that evaluates responsible riding behavior using speed compliance, smooth braking, and route adherence. For electric delivery vehicles, the platform provides real-time EV charging station availability, smart charging slot reservation, waiting-time prediction, and charging station recommendations integrated with delivery routes to minimize downtime.* *By combining artificial intelligence, predictive analytics, warehouse optimization, delivery safety monitoring, and EV charging management into a unified platform, the proposed solution improves warehouse productivity, enhances rider safety, reduces delivery delays, optimizes operational costs, supports sustainable transportation, and delivers a faster and more reliable customer experience.*
Keywords	Artificial Intelligence (AI), Smart Logistics, Quick Commerce, Warehouse Optimization, Indoor Navigation, Intelligent Order Batching, Route Optimization, Demand Forecasting, Delivery Partner Safety, Dynamic ETA Prediction, Electric Vehicle (EV), Smart Charging Management, Predictive Analytics, Inventory Optimization, Supply Chain Management.
Aprroval Status (for Guide)	YES
Meeting date & Time	16/07/2026
Student - Guide Interaction	

Guide Name	DR. PRIYA V
Guide Signature	
Approval Date	16/07/2026

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